Kaiyu Yang

kaiyuy@alumni.princeton.edu
yangky11.github.io · Google Scholar · github.com/yangky11

I lead a research team at Apodex working on large language models for mathematics and verification. AI systems generate code and proofs faster than anyone can check them; as generation gets cheaper, verification becomes the bottleneck. I have worked on machine learning for formal mathematics since my PhD.

Experience

Apodex, Redwood City, CA — Mar 2026 – Present
Lead Scientist

- Lead a small team of researchers on the mathematical capabilities of LLMs across the entire training stack: pre-training data that aims to capture all the math on the web, synthetic mid-training data distilled from research papers, and post-training pushing toward research-level mathematics.
- Build verifiers that help agents complete long-horizon tasks, applied to AI's own training pipelines, from the correctness of GPU kernels to the quality of training data.

Meta Fundamental AI Research (FAIR), New York, NY — Jun 2024 – Dec 2025
Research Scientist

- Advised Goedel-Prover and Goedel-Prover-V2 (COLM 2025, ICLR 2026), theorem provers that set the state of the art among open models at release, outperforming provers 20× their size; V2 is downloaded about 10,000 times a month.
- Co-developed ProofOptimizer (ICLR 2026), a model that simplifies machine-generated proofs without human demonstrations; collaborated on verified code generation (Verina, ICLR 2026; early adopters include AWS and Harmonic).

California Institute of Technology, Pasadena, CA — Sep 2022 – May 2024
Postdoctoral Fellow

- Built LeanDojo (NeurIPS 2023), the first open-source infrastructure for theorem proving with LLMs in Lean, used by projects at Meta, ByteDance, and DeepSeek.
- Autoformalizing Euclidean Geometry (ICML 2024) enabled the first faithful formalization of Euclid's *Elements*, Book I, in a modern proof assistant.

Education

Princeton University — PhD in Computer Science — 2022
University of Michigan — MS in Computer Science and Engineering — 2018
Tsinghua University — B.Eng. in Computer Science and B.S. in Mathematics — 2016

Selected Publications

Full list on Google Scholar.

- **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
  K. Yang, A. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, A. Anandkumar. NeurIPS 2023.
- **Formal Mathematical Reasoning: A New Frontier in AI**
  K. Yang, G. Poesia, J. He, W. Li, K. Lauter, S. Chaudhuri, D. Song. ICML 2025; Communications of the ACM.
- **Goedel-Prover-V2: Scaling Formal Theorem Proving with Scaffolded Data Synthesis and Self-Correction**
  Y. Lin, S. Tang, B. Lyu, Z. Yang, ..., **K. Yang**, ..., C. Jin. ICLR 2026.
- **Learning to Prove Theorems via Interacting with Proof Assistants**
  K. Yang, J. Deng. ICML 2019.

Open Source

Core developer of LeanDojo, ReProver, Lean Copilot, and CoqGym — open-source tooling for machine learning with the Lean and Coq proof assistants; 2,800+ GitHub stars across repositories.

Service and Community

- Co-organizer of the MATH-AI workshop at NeurIPS 2023, 2025, 2026; Area Chair for ICML, NeurIPS, and ECCV.
- Invited talks at Google, Meta FAIR, the Simons Institute, UC Berkeley, Stanford, and CMU; guest co-instructor of *Advanced Large Language Model Agents* (UC Berkeley & MOOC, 2025).
- Mentored students and junior researchers who went on to PhD programs at CMU, Toronto, Cornell, CUHK, and UIUC, and to industry labs; work covered in *Scientific American* (2024).